

Product Concept Document

Perfect Purifiers LLC | Product Development Planning | Written by Brian McCarthy

Market Problem / Opportunity

Perfect Purifiers LLC has an immediate opportunity to modernize its HEPA purifier line for customers who want cleaner, quieter, safer air purification without high maintenance burden. Current HEPA products are popular, but customers report unmet needs: limited VOC, virus, and odor removal; potential ozone generation; noisy operation; short filter life; expensive replacement filters; narrow room coverage; and maintenance issues that can allow mold and bacteria buildup. The market is attractive because the air purifier industry is projected to grow from USD 7.0B in 2023 to USD 13.4B by 2030, and PP LLC can leverage ten years of category experience and its existing distribution network.

Market Segments and Prioritization

Priority	Segment	Buyers / Users	Need and rationale
1 - Primary	Homeowners and households with 2+ adults, especially adults over age 35	Buyers: homeowners, heads of household, allergy-conscious consumers. Users: adults, children, guests, and pets in shared living spaces.	The scenario research shows 56% of home air purifier purchasers are above age 35, and 46% of homes with 2+ adults own an air purifier. This segment is purchase-ready and likely to value safety, low noise, larger coverage, and lower maintenance.
2 - Secondary	Families and health-conscious consumers in areas affected by smoke, seasonal allergens, odor, or hazardous air conditions	Buyers: parents, caregivers, apartment residents, and renters. Users: children, elderly family members, sensitive occupants, and remote workers.	This group has high need for broad contaminant reduction, odor management, no ozone emissions, and quiet operation during sleep and work. They may respond strongly to summer launch timing due to recent hazardous air concerns.
3 - Tertiary	Small offices, wellness spaces, and light commercial users	Buyers: office managers, clinics, studios, and facility leads. Users: employees, clients, and visitors.	Commercial buyers need more coverage, reliability, and lower operating cost. This segment can extend the market after the consumer line proves value, but it is not the first launch focus.

Initial Financial Projections and Market Window

Item	Projection / assumption	Implication
Conceive and Plan funding	\$30,000 initially allocated for team resource needs	Supports requirements definition, planning, concept approval, and business case approval.
Prototype cost	\$80,000 estimated to build three distinct prototypes with five core features	Enables Basic, Better, and Ultimate model validation before launch.
Expected returns	More than \$200,000 annually for two years; IRR expected above 200%	Justifies moving from Plan into development if risk controls remain acceptable.
Growth assumptions	Revenue increase of 25%, market expansion of 10%, product returns reduced by more than 50%	Provides measurable commercial targets for launch and post-launch review.
Market window	Launch within six months from January 5 for the summer hazardous-air season	Design January-February, development March-May, beta testing June, launch readiness by early July.

Competitive Landscape

Primary competitors include Donaldson Company Inc., Hengst, MANN+HUMMEL, Honeywell International Inc., and Daikin Industries Ltd. These competitors have scale, brand recognition, and existing filtration expertise, but PP LLC believes no single product line fully resolves the customer problems identified in the survey. PP LLC can differentiate through a tiered HEPA product family that combines lower noise, reduced or eliminated ozone, longer filter life, broader coverage, affordable replacement filters, and odor/cleaning functionality.

Features, Functionality, Logistics, and KPIs

Area	Planned content
Core features	Dual HEPA purifier with air freshener and cleaning option; noise reduction target of 50%; easy-remove lower-cost HEPA filter; filter life extended to 1.5 years; motor strength increased for 50% more coverage; ozone eliminated or significantly reduced.
Product options	Basic Purifier: reduced noise and ozone emissions. Better Purifier: reduced noise and ozone emissions plus stronger motor. Ultimate Purifier: stronger motor, longer-life HEPA filter, and air freshener/cleaning option.
Market logistics	Distribute through existing wholesale, retail, and online channels. Use PP LLC operations and partner channels to meet summer demand. Coordinate product positioning, pricing tiers, beta feedback, packaging, channel education, and return reduction.
Launch KPIs	Current versus post-launch market share, sales, revenue, customer satisfaction, product return rate, filter replacement cost perception, noise satisfaction, and percentage of beta issues resolved before launch.